$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; vector magnitude formula

+ dot product formula
prove via cosine law

Define $\vec{c} = \vec{a} \times \vec{b}$

$\Rightarrow |\vec{c}| = |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$

$= |\vec{a}||\vec{b}| \sqrt{1 - \cos^2\theta}$

$= \sqrt{|\vec{a}|^2|\vec{b}|^2 - |\vec{a}|^2|\vec{b}|^2 \cos^2\theta}$

$= \sqrt{(a_1^2+a_2^2+a_3^2)(b_1^2+b_2^2+b_3^2) - (a_1b_1+a_2b_2+a_3b_3)}$

$= \sqrt{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2}$ → symmetry

c_1? c_2? c_3?

$\Rightarrow$ assume $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ → $|\vec{c}| = \sqrt{c_1^2 + c_2^2 + c_3^2}$

$\Rightarrow$ to solve this issue, use $\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$.

$\Downarrow$

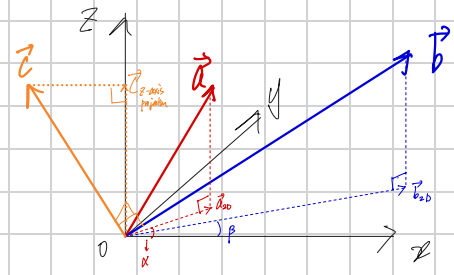
$a_1c_1 + a_2c_2 + a_3c_3 = 0$

$\Rightarrow$ $c_1 = \pm(a_2a_3 - a_3b_2)$

$c_2 = \pm(a_3b_1 - a_1b_3)$

$c_3 = \pm(a_1b_2 - a_2b_1)$

how to eliminate (±)?



$\vec{a}_{2D} = a_1\hat{i} + a_2\hat{j}$

$\vec{b}_{2D} = b_1\hat{i} + b_2\hat{j}$

$\Rightarrow \vec{c}_{z\text{-axis projection}}$ = one of $\pm(a_1b_2 - a_2b_1)\hat{k}$

$= c_3\hat{k} \Rightarrow$ find c_3

equal to $(|\vec{a}|\cos\alpha)\hat{i} + (|\vec{a}|\sin\alpha)\hat{j}$

equal to $(|\vec{b}|\cos\beta)\hat{i} + (|\vec{b}|\sin\beta)\hat{j}$

$\Rightarrow \pm(a_1b_2 - a_2b_1) = \pm\{(|\vec{a}|\cos\alpha)(|\vec{b}|\sin\beta) - (|\vec{a}|\sin\alpha)(|\vec{b}|\cos\beta)\}$

$= \pm|\vec{a}||\vec{b}|(\sin\beta\cos\alpha - \cos\beta\sin\alpha) = \pm|\vec{a}||\vec{b}|\sin(\beta - \alpha)$

<< case test >>

i) $0° < \beta - \alpha < 180° \Rightarrow c_3 > 0 \Rightarrow$ only $(a_1b_2 - a_2b_1) = |\vec{a}||\vec{b}|\sin(\beta - \alpha)$ satisfy

$\Downarrow$

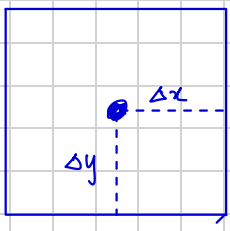
$c_3 = a_1b_2 - a_2b_1$

$\Downarrow$

similar procedures

$\Rightarrow$ also get c_1, c_2

<< Curl >>
2D



$F = P(x,y)\hat{i} + Q(x,y)\hat{j} \rightarrow \nabla \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ P & Q & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{k}$

torque (τ) 식 :

$\begin{cases} P(a, b-\Delta y) - P(a, b+\Delta y) \rightarrow \text{horizontal} \\ Q(a+\Delta x, b) - Q(a-\Delta x, b) \rightarrow \text{vertical} \end{cases}$

$\tau = \underbrace{r F \sin\theta \hat{k}}_{r \times F} \rightarrow \tau_{net} \approx \left(\{P(a, b-\Delta y) - P(a, b+\Delta y)\}\Delta x + \{Q(a+\Delta x, b) - Q(a-\Delta x, b)\}\Delta y \right)\hat{k}$

$\approx \left(\left\{ \frac{P(a, b+\Delta y) - P(a, b-\Delta y)}{-2\Delta y} 2\Delta y \right\}\Delta x + \left\{ \frac{Q(a+\Delta x, b) - Q(a-\Delta x, b)}{2\Delta x} \cdot 2\Delta x \right\}\Delta y \right)\hat{k}$

$\downarrow \lim_{\Delta \to 0}$

$\tau_{net} = \left(\left(-\frac{\partial P}{\partial y}\right) dy\,dx + \frac{\partial Q}{\partial x} dy\,dx \right)\hat{k}$

$= \left(\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dy\,dx \right)\hat{k}$

$\Rightarrow$ sign of τ_{net} comp. : $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$

$\Rightarrow \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$ is related to rotation